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The Reaserch on Parameters of Spectral Clustering Based on SVD

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***Abstract*—The spectral clustering algorithm, which cluster by using the eigenvalues and eigenvectors of Laplacian matrix, may not be obtained the desired clustering results in some cases. It is possible to remedy this deficiency by using the singular value decomposition (SVD) in the spectral clustering algorithm. It is presented in this article the algorithm of spectral clustering based on SVD which use singular value decomposition (SVD) instead of Laplacian eigenmaps. The choice of singular vectors, the estimate of the Gaussian kernel parameter, and the determine of the clusters number significantly influence the clustering results. The experiments show that these spectral clustering parameters such as the number of clusters, etc. can be calculated by use of the dimensionality reduction factor and the expression of singular value.**

Keywords*—*spectral clustering; singular value; singular vectors; singular value decomposition

I. INTRODUCTION

Clustering is the process of partitioning a set into k classes of similar objects. Each class, called cluster or block, consists of objects that are similar between themselves and dissimilar to others in some way. Namely, for a given data set $\boldsymbol{X} = \{x_1, x_2, ..., x_n\}$, we partition it into k classes $\boldsymbol{C}_1, \boldsymbol{C}_2, ..., \boldsymbol{C}_k$, $\boldsymbol{C}_i \subseteq \boldsymbol{X}$ and satisfies the following condition [1]:

$$\bigcup_{i=1}^{k} \boldsymbol{C}_i = \boldsymbol{X}, \forall i \neq j, \boldsymbol{C}_i \bigcap \boldsymbol{C}_j = \emptyset, \boldsymbol{C}_i \neq \emptyset. \quad (1)$$

Compared with most of clustering algorithms, the spectral clustering has ability to cluster on the sample space of arbitrary shape, and converge to the global optimal solution [2].

The clustering results of the spectral clustering algorithm that cluster by using the eigenvectors of Laplacian matrix, in some cases, are not satisfactory.

The Laplacian matrix, which is a similarity matrix after Laplacian eigenmaps, has the property of its eigenvalues greater than or equal to 0, i.e. the matrix is positive semi-definite.

For a similarity matrix $\boldsymbol{A}$, the matrix $\boldsymbol{A}\boldsymbol{A}^T$ is positive semi-definite. Therefore, in the spectral clustering algorithm, we can use the singular value decomposition (SVD) instead of the Laplacian eigenmaps.

In this paper, we study the application of singular value decomposition in spectral clustering, and discuss the approach of determining some parameters in the algorithm of spectral clustering based on SVD.

For the sake of discussion, in this paper, the spectral clustering algorithm that cluster by using the eigenvectors of Laplacian matrix is called as common spectral clustering algorithm.

II. SPECTRAL CLUSTERING

The common spectral clustering has some different implement algorithms [2], [3], [5], [9], which can be briefly stated as follows [3]:

Given a data set $\boldsymbol{X} = \{x_1, x_2, ..., x_n\}, x_i \in \Re^N$. Construct a weighted graph $\boldsymbol{G} = (\boldsymbol{V}, \boldsymbol{E})$, wherein $\boldsymbol{V} = \{v_i, i = 1, 2, ..., n\}$ is the set of vertices, $\boldsymbol{E} = \{e_{ij}, i = 1, 2, ..., n, j = 1, 2, ..., n\}$ is the set of edges that connect vertices(v_i, v_j). Each vertex v_i of the graph corresponds to a point x_i.

(1)Construct the similarity matrix $\boldsymbol{W}$ of data set $\boldsymbol{X}$ by using a similarity rule, and calculate the non-normalized Laplacian matrix $\boldsymbol{L}$;

(2)Calculate the eigenvalues and eigenvectors of Laplacian matrix $\boldsymbol{L}$, and select the first k eigenvectors $\boldsymbol{u}_1, \boldsymbol{u}_2, ..., \boldsymbol{u}_k$;

(3)Construct the matrix $\boldsymbol{V} \in \Re^{n \times k}$ with the eigenvectors $\boldsymbol{u}_1, \boldsymbol{u}_2, ..., \boldsymbol{u}_k$ as columns, interpret the rows of $\boldsymbol{V}$ as new vectors $\boldsymbol{z}_i \in \Re^k, i = 1, 2, ..., n$;

(4)Cluster the vectors of $\boldsymbol{z}_1, \boldsymbol{z}_2, ..., \boldsymbol{z}_n$ with k-means clustering algorithm into k clusters $\boldsymbol{C}_1, \boldsymbol{C}_2, ..., \boldsymbol{C}_k$;

(5)According to the corresponding relationship between x_i and $\boldsymbol{z}_i$, determine the clustering results of the data set $\boldsymbol{X}$.

In the common spectral clustering algorithm, it is necessary to perform Laplacian eigenmaps with the similarity matrix $\boldsymbol{W}$ so that the eigenvalues of Laplacian matrix greater than or equal to 0, i.e. the Laplacian matrix is positive semi-definite.

The $n \times n$ non-normalized Laplacian matrix $\boldsymbol{L}$ of similarity matrix $\boldsymbol{W}$ is defined by

$$\boldsymbol{L} = \boldsymbol{D} - \boldsymbol{W} \quad (2)$$

Wherein $\boldsymbol{D}$ is the $n \times n$ diagonal matrix with elements $d_{ii} = \sum_{j=1}^{n} w_{ij}$, $i = 1, 2, ..., n$ [4].

III. SINGULAR VALUE DECOMPOSITION

Singular value decomposition, or SVD, is an important method of matrix factorization in linear algebra, which has been widely used in dimensionality reduction of pattern recognition, solution of the matrix rank, as well as information retrieval [6].

Definition 1: Let $\boldsymbol{A}$ be a $m \times n$ matrix, the eigenvalues of $\boldsymbol{A}^T \boldsymbol{A}$ are

$$\lambda_1 \geqslant \lambda_2 \geqslant ... \geqslant \lambda_r \geqslant \lambda_{r+1} = ... = \lambda_n = 0 \tag{3}$$

then $\sigma_i = \sqrt{\lambda_i}(i = 1, ..., r)$ are called the singular values of matrix $\boldsymbol{A}$.

Theorem 1: Let $\boldsymbol{A}$ be a $m \times n$ real matrix, its rank is r, then there exists a $m \times m$ orthogonal matrix $\boldsymbol{U}$, a $n \times n$ orthogonal matrix $\boldsymbol{V}$ and a $m \times n$ diagonal matrix $\boldsymbol{\Sigma}$, so that

$$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^T \tag{4}$$

wherein the diagonal matrix $\boldsymbol{\Sigma} = diag(\sigma_1, \sigma_2, ..., \sigma_r, 0, ..., 0)$, σ_i are the singular values of matrix $\boldsymbol{A}$ and satisfy $\sigma_1 \geqslant \sigma_2 \geqslant ... \geqslant \sigma_r$.

The equation (4) can be expressed as

$$\boldsymbol{A} = \sum_{i=1}^{r} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i{}^T \tag{5}$$

the vectors $\boldsymbol{u}_i, \boldsymbol{v}_i$, which are column vectors of the matrix $\boldsymbol{U}$, $\boldsymbol{V}$, are called left and right singular vectors of $\boldsymbol{A}$ respectively.

The singular value of matrix $\boldsymbol{A}$ can be divided into two kinds: The principal singular value – k large singular value, and secondary singular value – the singular value of other smaller.

It is important that choose the number of the principal singular value k. On the one hand, k should be large enough to reflect the information and structure of the original data; on the other hand, k should be small enough, so that filter out all the irrelevant and redundant information and details.

IV. Spectral Clustering Based on SVD

In this section we discuss the algorithm of spectral clustering based on SVD, as well as the approach to determine some parameters in the algorithm.

A. The algorithm of spectral clustering based on SVD

The algorithm of spectral clustering based on SVD is:

Input: Given a data set $\boldsymbol{X} = \{x_1, x_2, ..., x_n\}, x_i \in \Re^N$.

Step 1 By use of the distance $d(x_i, x_j)$ between objects x_i and x_j calculate the similarity $s(x_i, x_j)$ [4], [6]

$$s(x_i, x_j) = exp\left(-d(x_i, x_j)^2 / 2\sigma^2\right) \tag{6}$$

According to the similarity of objects, construct the similarity matrix $\boldsymbol{A}$ of data set $\boldsymbol{X}$;

Step 2 Calculate the singular value decomposition of the similarity matrix $\boldsymbol{A}$ [4]:

$$[\boldsymbol{u} \;\; \boldsymbol{s} \;\; \boldsymbol{v}] = svd(\boldsymbol{A}) \tag{7}$$

Wherein, $\boldsymbol{u}$ and $\boldsymbol{v}$ are left and right singular vector of $\boldsymbol{A}$ respectively, $\boldsymbol{s} = diag(\sigma_1, \sigma_2, ..., \sigma_n)$, σ_i are the singular value of $\boldsymbol{A}$.

Step 3 Select the first l vectors $\boldsymbol{u}_1, \boldsymbol{u}_2, ..., \boldsymbol{u}_l$ of the left singular vector matrix $\boldsymbol{u}$, and construct a matrix $\boldsymbol{U} \in \Re^{n \times l}$ with the vectors as columns;

Step 4 Let $y_i \in \Re^l (i = 1, 2, ..., n)$ be the vector corresponding to the row i of matrix $\boldsymbol{U}$. Cluster the vector y_i with the k-means clustering algorithm into k clusters $\boldsymbol{C}_1, \boldsymbol{C}_2, ..., \boldsymbol{C}_k$.

Step 5 Construct a mapping

$$x_i \in \Re^N \mapsto y_i \in \Re^l, i = 1, 2, ..., n \tag{8}$$

According to the corresponding relationship between x_i and y_i determine the clustering results of data set $\boldsymbol{X}$.

Output: Output the clustering results of data set $\boldsymbol{X}$.

In step 3, we choose the first l column vectors $\boldsymbol{u}_1, \boldsymbol{u}_2, ..., \boldsymbol{u}_l$ of the left singular matrix $\boldsymbol{u}$, of course, we can also choose the first l row vector $\boldsymbol{v}_1, \boldsymbol{v}_2, ..., \boldsymbol{v}_l$ of the right singular matrix $\boldsymbol{v}$.

In the algorithm of spectral clustering based on SVD, there are three parameters: k, θ and l, the values of these parameters have an important impact on the results of clustering.

B. Computational complexity

In the algorithm of spectral clustering based on SVD, we have to construct the similarity matrix of between vertices of the graph $\boldsymbol{G}$, calculate the singular values and singular vectors of the similarity matrix. When constructing the similarity matrix, the space complexity is $O(n^2)$, and the time complexity is $O(n^3)$. When calculating singular values and singular vectors, the time complexity is $O(n^3)$. Therefore, in the algorithm of spectral clustering based on SVD, the space complexity is $O(n^2)$, the time complexity is $O(n^3)$.

As with the common spectral clustering algorithm, these is a bottleneck problem in the algorithm of spectral clustering based on SVD that has high space complexity and time complexity in practical applications.

C. Determining the number of clusters k

These are two kinds of methods to calculate the number of clusters k as follows: One is by use of the expression of singular value, and another by use of the expression of dimensionality reduction factor.

1) Using the expression of singular value to determine the number of clusters: We assume that the number of clusters k of spectral clustering based on SVD and the number of the principal singular value of the similarity matrix $\boldsymbol{A}$ are equivalent. The expression of singular value can be applied to determine the number of clusters k.

The literature [3], [10] proposed that the number of clusters in common spectral clustering can be determined by using the subscript k of the maximum of eigenvalues difference sequence $\lambda_k - \lambda_{k+1}$, wherein the similarity s in algorithm is expressed with Gaussian kernel function (6).

In the general case, reference to the common spectral clustering algorithm, we can determine the number of clusters by use of the subscript k of the maximum of singular value difference sequence $\sigma_k - \sigma_{k+1}$. However, in some cases, it is not accurate. While the number of clusters, which determined by using the subscript k of the maximum of singular value sequence expression $\sigma_k - 2\sigma_{k+1} + \sigma_{k+2}$, is more accurate relatively. Therefore, the number of clusters k can be derived from the following formula:

$$\arg\max_{k} \{\sigma_k - 2\sigma_{k+1} + \sigma_{k+2} | k > 1\} \tag{9}$$

2) Using dimensionality reduction factor to determine the number of clusters: From the perspective of dimensionality reduction, the clustering problem can be seen as that the n-dimensional data sets $\boldsymbol{X}$ is expressed by k low-dimensional clusters $\boldsymbol{C}_i$, i.e. $\boldsymbol{X} = \bigcup_{i=1}^{k} \boldsymbol{C}_i$. The number of clusters can be determined by using dimension reduction factor. That is, for a given constant $\theta(0 < \theta < 1)$, the clusters number k of spectral clustering based on SVD can be derived from the following formula:

$$\arg\min_k \left\{ k \middle| \sum_{i=1}^{k} \sigma_i \middle/ \sum_{i=1}^{n} \sigma_i \geqslant \theta \right\} \tag{10}$$

Usually the case, the θ value can be derived through the experiment. For example, when the similarity s is expressed with Gaussian kernel function (6), the parameter θ is 0.850.

D. Determining the parameter σ

Due to the singular value associated indirectly with the parameter σ in the Gaussian kernel function, therefore, the number of clusters k is affected by the parameter σ.

Adjust the parameter σ in Gaussian kernel function (6), the similarity between objects will be changed, and the singular value σ_i of similarity matrix will be changed also.

When the number of clusters that determined respectively by using the equation (9) and (10) are consistent, we will obtain the suitable Gaussian kernel parameter σ and the number of clusters k.

E. Determining the parameter l

In the algorithm of spectral clustering based on SVD, the singular vector $\boldsymbol{u}_1, \boldsymbol{u}_2, ...\boldsymbol{u}_l$ which is chosen from the front l vectors of the matrix $\boldsymbol{u}$ significantly influence the clustering results. Under normal circumstances, we first choose $l = k$. If the clustering results are not satisfactory, by adjusting the number l of singular vectors, we may be able to obtain the desired clustering results.

V. Experimental Results and Analysis

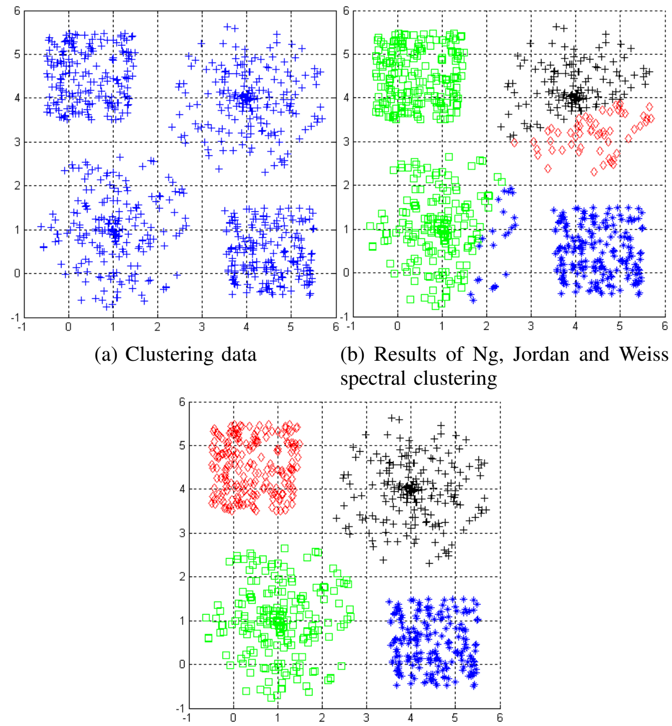
In order to verify the correctness of the algorithm of spectral clustering based on SVD, as well as the effectiveness of the methods for determining some parameters, we select three experiments in the following section.

The first experiment is to compare the clustering results between the common spectral clustering and the spectral clustering based on SVD; the second to verify the effectiveness of selection method for parameters σ and k; the third to verify the effectiveness of selection method for the first l singular vector $\boldsymbol{u}_1, \boldsymbol{u}_2, ...\boldsymbol{u}_l$.

The following experiments are performed under MATLAB 7.5, the results of singular value decomposition are achieved by use of the $svd()$ function in MATLAB.

A. Experiment 1

For the data in Fig.1(a), we use the Ng, Jordan and Weiss spectral clustering and the spectral clustering based on SVD for clustering respectively. Wherein, the number of clusters is 4, the parameter σ of Ng, Jordan and Weiss spectral clustering is 1.98, the parameter σ of spectral clustering based on SVD is 2.0. The clustering results are shown in the Fig.1(b) and Fig.1(c) respectively.



(a) Clustering data

(b) Results of Ng, Jordan and Weiss spectral clustering

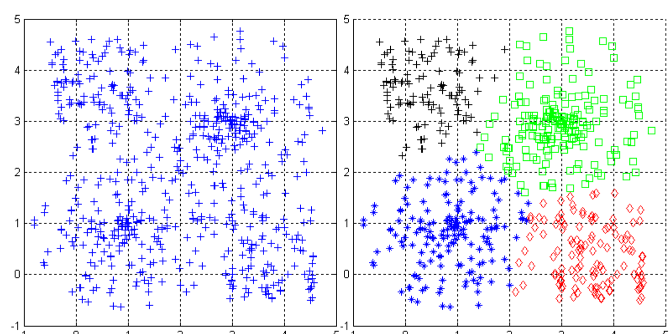
(c) Results of spectral clustering based on SVD

Fig. 1. The results of experiment 1

It can be seen from Fig.1(b) and Fig.1(c), for the data in Fig.1(a), the clustering results of the Ng, Jordan and Weiss spectral clustering are not satisfactory, while by using the spectral clustering based on SVD we have got the desired clustering results.

B. Experiment 2

For the data in Fig.2(a), we determine the number of clusters k and parameter σ by using equation (9) and (10).



(a) Clustering data

(b) Clustering results

Fig. 2. The results of experiment 2

First, we choose the Gauss kernel parameter σ =1, the values of the singular value sequence expression σ_k, $\sigma_k - \sigma_{k+1}$, $\sigma_k - 2\sigma_{k+1} + \sigma_{k+2}$ and $\sum_{i=1}^{k} \sigma_i / \sum_{i=1}^{n} \sigma_i$ are shown in table 1 (wherein, $\sum_{i=1}^{n} \sigma_i$ =600).

When k = 4, the singular value sequence expression $\sigma_k - \sigma_{k+1}$ and $\sigma_k - 2\sigma_{k+1} + \sigma_{k+2}$ has been reached the maximum (in the case of $k > 1$).

When $k \geqslant 10$, $\sum_{i=1}^{k} \sigma_i / \sum_{i=1}^{n} \sigma_i \geqslant \theta$ (θ=0.850).

These number of clusters that are determined respectively by use of equation (9) and (10) are 4 and 10. These two number of clusters are inconsistent, the Gaussian kernel parameter σ = 1 is not the best value.

TABLE I. SMALL CAPSPART OF SINGULAR VALUE WHEN σ = 1

k	σ_k	$\sigma_k - \sigma_{k+1}$	$\sigma_k - 2\sigma_{k+1} + \sigma_{k+2}$	$\sum_{i=1}^{k} \sigma_i \Big/ \sum_{i=1}^{n} \sigma_i$
1	128.4	30.0	7.2	0.2141
2	98.4	22.9	13.2	0.3781
3	75.6	9.7	-19.6	0.5040
4	65.9	29.2	21.8	0.6138
5	36.6	7.4	6.2	0.6749
6	29.2	1.2	-6.8	0.7236
7	28.0	8.1	5.2	0.7703
8	19.9	2.9	1.4	0.8035
9	17.0	1.5	1.0	0.8318
10	15.5	0.6	-3.3	0.8576

Adjust the Gaussian kernel parameter σ, when σ = 1.7, the value of the singular value sequence expression σ_k, $\sigma_k - \sigma_{k+1}$, $\sigma_k - 2\sigma_{k+1} + \sigma_{k+2}$ and $\sum_{i=1}^{k} \sigma_i / \sum_{i=1}^{n} \sigma_i$ are shown in table 2 (wherein, $\sum_{i=1}^{n} \sigma_i$ =600).

TABLE II. SMALL CAPSPART OF SINGULAR VALUE WHEN σ = 1.7

k	σ_k	$\sigma_k - \sigma_{k+1}$	$\sigma_k - 2\sigma_{k+1} + \sigma_{k+2}$	$\sum_{i=1}^{k} \sigma_i \Big/ \sum_{i=1}^{n} \sigma_i$
1	246.6	136.9	128.7	0.4110
2	109.7	8.2	-38.3	0.5938
3	101.5	46.5	15.3	0.7630
4	55.0	31.2	26.6	0.8546
5	23.8	4.6	-1.1	0.8943
6	19.2	5.7	0.3	0.9263
7	13.5	5.4	3.3	0.9488
8	8.1	2.1	1.1	0.9622
9	5.9	1.0	-0.3	0.9721
10	4.9	1.3	0.1	0.9803

When k = 4, the value of the singular value sequence expression $\sigma_k - 2\sigma_{k+1} + \sigma_{k+2}$ has been reached a maximum (in the case of $k > 1$), $\sum_{i=1}^{k} \sigma_i / \sum_{i=1}^{n} \sigma_i \geqslant \theta$ (θ=0.850).

These number of clusters that determined respectively by using equation (9) and (10) are consistent. We have got the best Gaussian kernel parameters σ = 1.7, and the best number of clusters k = 4.

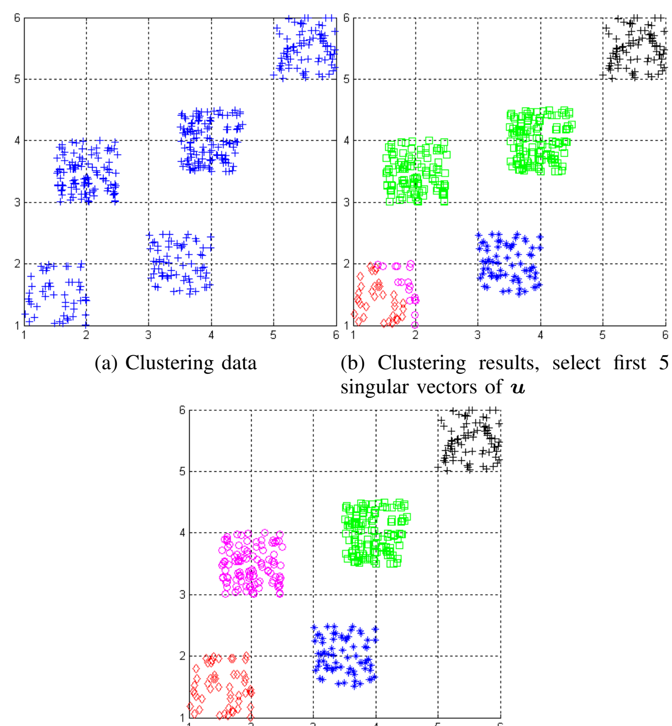
For the data in Fig.2(a), the clustering results that use the spectral clustering based on SVD are shown in the Fig.2(b), wherein the parameter σ = 1.7, k = 4 and l = 4.

As can be seen from table 2, in this issue, when k = 3, the value of the singular value sequence expression $\sigma_k - \sigma_{k+1}$ has been reached a maximum (in the case of $k > 1$), the subscript k is inconsistent with the best clusters number (k = 4). The number of clusters that determined by use of the subscript k of the maximum of sequence $\sigma_k - 2\sigma_{k+1} + \sigma_{k+2}$ (in the case of $k > 1$) is more accurate.

C. Experiment 3

This experiment is about how to select the first l singular vector $\boldsymbol{u}_1, \boldsymbol{u}_2, ..., \boldsymbol{u}_l$ of the matrix $\boldsymbol{u}$.

Usually the case, for the number of singular vectors, l, we should first choose $l = k$ (the number of clusters). When the clustering results are not satisfactory, we may adjust the value of l to achieve the optimal clustering results.



(a) Clustering data

(b) Clustering results, select first 5 singular vectors of $\boldsymbol{u}$

(c) Clustering results, select first 4 singular vectors of $\boldsymbol{u}$

Fig. 3. The results of experiment 3

For the data in Fig.3(a), the best choice of parameters σ and k are 0.96 and 5 respectively. First at all, we choose $l = k = 5$, the clustering results are shown in the Fig.3(b). As can be seen from Fig.3(b), the clustering results are not satisfactory. When l = 4, we have got a satisfactory clustering results as shown in the Fig.3(c).

VI. CONCLUSION

For the purpose of making the eigenvalues of similarity matrix greater than or equal to 0, i.e. similarity matrix is positive semi-definite, it is necessary to perform Laplacian eigenmaps with the similarity matrix in the algorithm of common spectral clustering.

Since use the eigenvectors of the Laplacian matrix $\boldsymbol{L}$ rather than that of similarity matrix $\boldsymbol{A}$, we can't get the desired clustering results in the algorithm of common spectral clustering in some cases.

In virtue of using the matrix $\boldsymbol{A}\boldsymbol{A}^T$ instead of the Laplacian matrix $\boldsymbol{L}$, the spectral clustering based on SVD, in the aspects of clustering, is more effective than the common spectral clustering.

The similarity of these objects with in a class, between different classes and with other classes can be very close to each other, therefore, two classes of objects whose properties are completely different or opposite may be clustered into one class. The choice of singular vectors, the estimate of the Gaussian kernel parameter and the determine methods of the clusters number significantly influences the results of clustering.

We can derived some methods from equation (9) and (10), such as the determine of clusters number k and the estimate of Gaussian kernel parameter σ as well as the choice of the singular vectors.

As with other spectral clustering algorithm, these is a bottleneck problem in practical application that has high space complexity and time complexity in the spectral clustering based on SVD. Is is a problem to be solved that reduce the space complexity and time complexity of spectral clustering algorithm in the future research.

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